Game Design Document (GDD) for Obstacle Assault

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Summary:

Obstacle Assault is a 3D precision platformer built entirely in Unreal Engine. The game strips away combat, narrative, and UI hand-holding to focus purely on the mathematics of movement. The core design pillar is simple: **Ensure players feel challenged by the environment, never by the controls.** The experience relies on spatial reasoning, timing, and an instant-restart loop to keep players in a continuous Flow State.

Game References: *Super Meat Boy* (Pacing/Restarts), *Celeste* (Movement Forgiveness), *Stumble Guys* (Platform Chaos).

The 3Cs (Character, Camera, Controls)

To eliminate "floaty" platforming and cheap deaths, the movement logic was heavily customized away from standard engine physics.

- **Character Physics:**
  - **Ground Friction:** Drastically increased. The character must snap to a halt the exact frame directional input ceases to prevent edge-slipping.
  - **Gravity Scale (Descent):** Multiplied by 1.5x after reaching the jump apex. This creates a fast, weighty fall, giving the player immediate ground feedback.
- **Control Forgiveness (Hidden Assists):**
  - **Coyote Time (0.15s):** Allows a valid jump input slightly *after* the character has run off a ledge, forgiving human reaction limits.
  - **Input Buffering (0.1s):** Remembers a jump input pressed slightly *before* hitting the ground, executing it perfectly upon landing.
- **Camera Configuration:**
  - **Arm Length:** 450uu. Positioned slightly higher than a standard third-person camera to grant maximum visibility of the landing zone.

Spatial Metrics & The Grid

You cannot design a fair obstacle course without mathematical limits. Every gap and hazard in the game is derived from these three numbers:

- **Standard Jump Distance:** Max 400uu (Unreal Units)
- **Sprint Jump Distance:** Max 650uu
- **Vertical Apex:** Max 250uu
- *Design Rule:* A gap of 600uu represents a "High Tension" jump. A gap of 300uu represents a "Traversal/Pacing" jump.

Level Pacing & Invisible Tutorials

The game uses level architecture to teach mechanics safely before requiring the player to perform them under pressure. No pop-up text is used.

1. **Prologue (The Tutorial):** Introduce hazard in safety (over solid ground). Failure = try again.
2. **Chapter 1 (The Escalation):** Test with consequence (over a void). Failure = instant restart.
3. **Chapter 2 (The Climax):** Combine hazards (e.g., moving + rotating). The toughest combinations testing everything learned, ending in a deliberate moment of calm.

Platform Psychology & Tension Values

The type of platform chosen shifts the required skill from raw timing to spatial adaptability. I assigned emotional states and difficulty metrics to individual obstacles to control the pacing arc.

- **Idle / Safe Zone (Tension: 1)**
  - *Why:* Provides a breather between chaotic sequences.
  - *Feels Like:* Relief, anticipation.
- **Up & Down (Tension: 2)**
  - *Why:* Teaches vertical timing early.
  - *Feels Like:* Urgency, rhythm focus.
- **Rotating (Tension: 3)**
  - *Why:* Challenges spatial control and patience.
  - *Feels Like:* Careful, constant visual adjustment.
- **Pop In & Out (Tension: 4)**
  - *Why:* Forces jump commitment without hesitation.
  - *Feels Like:* Intense timing pressure.
- **Floating Over Thorns (Tension: 5 - Lethal)**
  - *Why:* One slip means game over. Requires high-stakes precision.
  - *Feels Like:* Pure fear of failure.

Hazard Combinations

Combining base archetypes multiplies the Tension Level, reserved strictly for Chapter 2.

- **Pop In & Out + Thorns:** The apex of difficulty. The player must commit to a jump toward a platform that currently doesn't exist, threading the needle over a lethal reset zone.

The Instant Restart System

- **Mechanism:** Upon triggering a hazard, character location resets to the last visible checkpoint in < 0.5 seconds.
- **Design Intent:** By reducing the failure penalty to zero load time, players are encouraged to experiment with risky, high-speed routes without the fear of wasting their time.